

This manual is for creating a ssh server on Jetson Nano using tailscale VPN. [More on tailscale here.](#)

Prerequisite:

1. Properly flashed SD card with the most recent jetpack version
2. Ethernet or WiFi connection
3. Admin permission on the Jetson's Ubuntu GUI
4. keyboard & mouse
5. Monitor with HDMI or DVI connection
6. An account with tailscale and log in on the chromium browser on the Jetson
7. Nano text editor

Step-1: Opening a free account with tailscale

Please click [Tailscale](#) and follow their instruction to create your free account

Step-2: Setting up the environment

It's a good practice to always update and upgrade everything installed on the system before installing or setting up new dependencies.

Run

```
sudo apt-get update
```

```
sudo apt-get upgrade -y
```

Step-3: Installing dependencies on the machine

Run:

```
sudo apt install -y curl gnupg openssh-server
```

Step-4: Installing Tailscale

Run:

```
curl -fsSL https://tailscale.com/install.sh | sh
```

Step-5: Enable ssh

Run:

```
sudo systemctl enable --now ssh
```

Step-6: Building Tailscale on the machine

Run:

```
sudo tailscale up
```

This will return a link. Copy this link and paste it on the browser. Upon pressing enter, the link will take to a page where it will ask to connect the machine to the network. Upon clicking connect, the machine will be visible on admin console page.

Step-7: Modifying sshd config

Run:

```
sudo nano /etc/ssh/sshd_config
```

add these lines in the file

PasswordAuthentication yes

PermitRootLogin no

PubkeyAuthentication yes

Press Ctrl+O, then press Enter, then press Ctrl+X to save and exit

Step-8: Keygen for ssh

Run:

```
sudo ssh-keygen -A
```

Step-9: Restarting ssh

Run:

```
sudo systemctl restart ssh
```

Step-10: Setting hostname

Run:

```
sudo tailscale up --reset --ssh=false --hostname PLACEHOLDER
```

replace PLACEHOLDER with the hostname you want to use. If the system username is X and the hostname is Y-desktop, set the hostname to Y for simplicity.

Step-11: Checking if ssh is running

Run:

```
sudo service ssh status
```

Look for “Active: **active (running)**”

If it's exactly like that, the ssh server has successfully been installed and it's running

Step-12: Checking on local host

On Jetson's terminal, run:

```
ssh X@localhost
```

If the setup has been successful, the terminal will ask for input in the form of yes/no for authentication (first time login from any device and local host). Type “yes” and enter the password.

The next page shows how to setup client devices.

Setting Up Client device

Setting up client devices: Windows

Step-1: Opening a free account with tailscale and stay logged in on the client machine

Step-2: [Download the latest tailscale .exe file from this link](#)

Step-3: Login using the same account and connect the machine.

Step-4: Ask for an invitation link or email from the owner or admin of the ssh server. Click on the provided link and connect.

Step-5: Open a powershell terminal with admin privilege.

Run:

```
ssh X@ip -p 22
```

Where X comes from X@Y-desktop

ip comes from the tailscale network both machines are connected

Setting up Client Devices: Ubuntu

Step-1: [Follow this tutorial to get started on Linux](#)

Step-2: Ask for an invitation link or email from the owner or admin of the ssh server. Click on the provided link and connect.